

19ECE204
Digital Electronics and Systems
III Semester
B. Tech. (ECE & CSE)
Amrita School of Engineering,
Bengaluru
Lecture 45 to 46

Steps to Design a Synchronous Counter

For synchronous counters, all the flip-flops are using the same CLOCK signal. Thus, the output would change synchronously.

Procedure to design synchronous counter are as follows:-

STEP 1: Obtain the [State Diagram](#).

STEP 2: Obtain the [Excitation Table](#) using state transition table for any particular FF (JK or D). Determine number of FF used.

STEP 3: Obtain and simplify the function of each FF input using [K-Map](#).

STEP 4: Draw the [circuit](#).

Excitation Tables

During the design of circuits, we need a table that lists the required input combinations for a given change of state. Such table is called a flip flop *excitation table*.

RS	
Q_i	Q_{i+1}
0	0
0	1
1	0
1	1

JK	
Q_i	Q_{i+1}
0	0
0	1
1	0
1	1

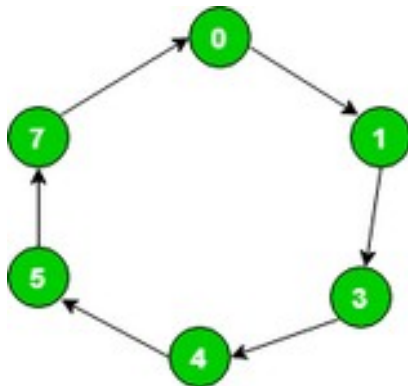
D	
Q_i	Q_{i+1}
0	0
0	1
1	0
1	1

T	
Q_i	Q_{i+1}
0	0
0	1
1	0
1	1

- Design synchronous counter for sequence: $0 \rightarrow 1 \rightarrow 3 \rightarrow 4 \rightarrow 5 \rightarrow 7 \rightarrow 0$, using T flip-flop.
- For given sequence, state transition diagram as following below

State transition table logic:

PRESENT STATE	NEXT STATE
0	1
1	3
3	4
4	5
5	7
7	0



PRESENT STATE			NEXT STATE		
Q_3	Q_2	Q_1	$Q_3(t+1)$	$Q_2(t+1)$	$Q_1(t+1)$
0	0	0	0	0	1
0	0	1	0	1	1
0	1	1	1	0	0
1	0	0	1	0	1
1	0	1	1	1	1
1	1	1	0	0	0

INPUT TABLE OF FLIP-FLOPS			
T_3	T_2	T_1	
0	0	1	
0	1	0	
1	1	1	
0	0	1	
0	1	0	
1	1	1	

T flip-flop – If value of Q changes either from 0 to 1 or from 1 to 0 then input for T flip-flop is 1 else input value is 0.

Q_T	Q_{T+1}	T
0	0	0
0	1	1
1	0	1
1	1	0

Compare Q_n with $Q_n(t+1)$ and use excitation table of the flipflop to be used and write the excitation table for the counter to be designed

Find value of T_3 , T_2 , T_1 in terms of Q_3 , Q_2 , Q_1 using K-Map (Karnaugh Map):

T3 $Q_3 \backslash Q_2 Q_1$	00	01	11	10
0	0	0	1	X
1	0	0	1	X

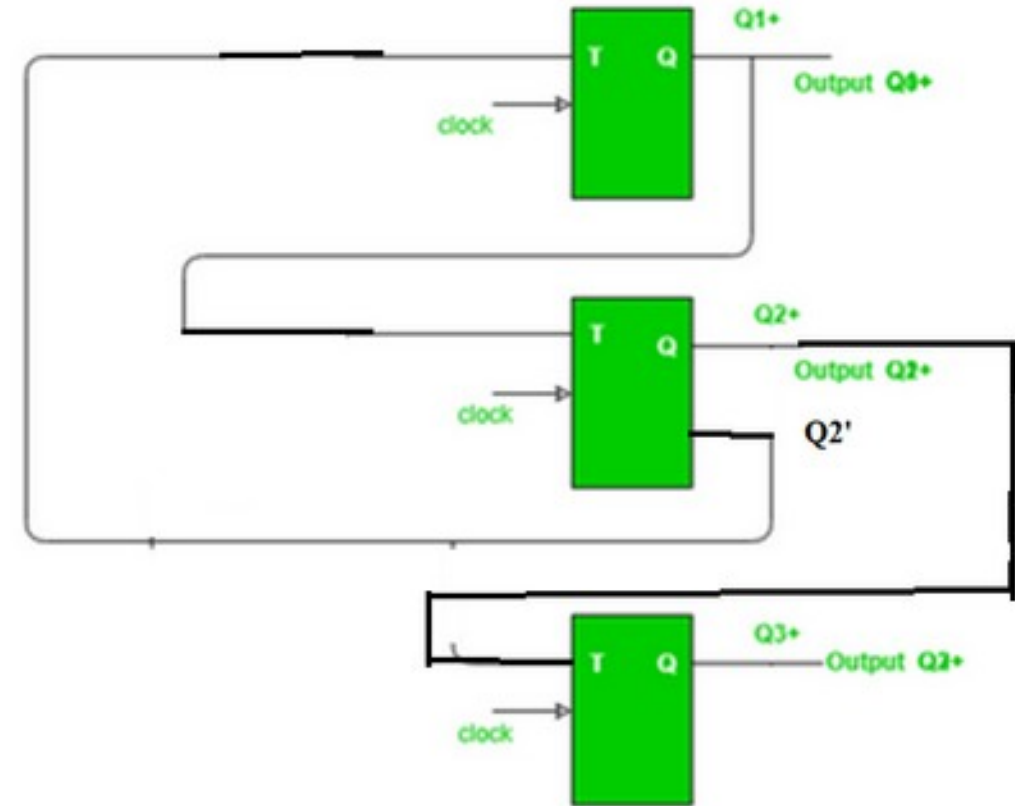
Therefore,
 $T_3 = Q_2$

T2 $Q_3 \backslash Q_2 Q_1$	00	01	11	10
0	0	1	1	X
1	0	1	1	X

Therefore,
 $T_2 = Q_1$

T1 $Q_3 \backslash Q_2 Q_1$	00	01	11	10
0	1	1	0	X
1	1	1	0	X

Therefore,
 $T_1 = Q_2'$



Design a Synchronous Counter to Count 4,7,3,0 and 2 respectively using JKFlip Flop negative triggered by showing:

- i. Flip Flop Used
- ii. State Transition Diagram
- iii. Excitation Table / Present state, next State
- iv. Karnough Map & perform Simplified Function
- v. The Synchronous Counter

Solution:

Step 1 : Flip Flop Used

Find Modulo, $N = 2^n$

$$M = 7, M = 2^n - 1 = 7$$

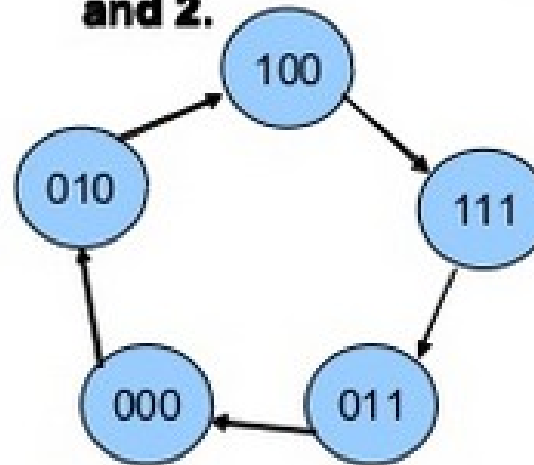
$$2^n = N, \text{ so, } N = 7 + 1 = 8, \text{ MOD } 8$$

$$2^n = 8, n = \log 8 / \log 2$$

$n = 3 \text{ bit} = 3 \text{ Flip Flop.}$

Solution:

Step 2 : State Transition Diagram, to count 4, 7, 3, 0 and 2.



Excitation Truth Table For Counter using JK FlipFlop

Present Q_n	Next State Q_{n+1}	J	K
0	0	0	x
0	1	1	x
1	0	x	1
1	1	x	0

Step 3 : Exitation Table /present State, Next State

Dec mal	Present State			Next State			JC	KC	JB	KB	JA	KA
	QC	QB	QA	QC	QB	QA						
4	1	0	0	1	1	1	x	0	1	x	1	x
7	1	1	1	0	1	1	x	1	x	0	x	0
3	0	1	1	0	0	0	0	x	x	1	x	1
0	0	0	0	0	1	0	0	x	1	x	0	x
2	0	1	0	1	0	0	1	x	x	1	0	x

K-Map For
JA = QC

A \ CB	00	01	11	10
0	0	0	x	1
1	x	x	x	x

K-Map For
KA = QC

A \ CB	00	01	11	10
0	x	x	x	x
1	x	1	0	x

Step 3 : Exitation Table /present State, Next State

Deci mal	Present State			Next State			JC	KC	JB	KB	JA	KA
	QC	QB	QA	QC	QB	QA						
4	1	0	0	1	1	1	x	0	1	x	1	x
7	1	1	1	0	1	1	x	1	x	0	x	0
3	0	1	1	0	0	0	0	x	x	1	x	1
0	0	0	0	0	1	0	0	x	1	x	0	x
2	0	1	0	1	0	0	1	x	x	1	0	x

K-Map For
JB = 1

A \ CB	00	01	11	10
0	1	x	x	1
1	x	x	x	x

K-Map For
KB = QC

A \ CB	00	01	11	10
0	x	1	x	x
1	x	1	0	x

K-Map For
JC = QA'QB

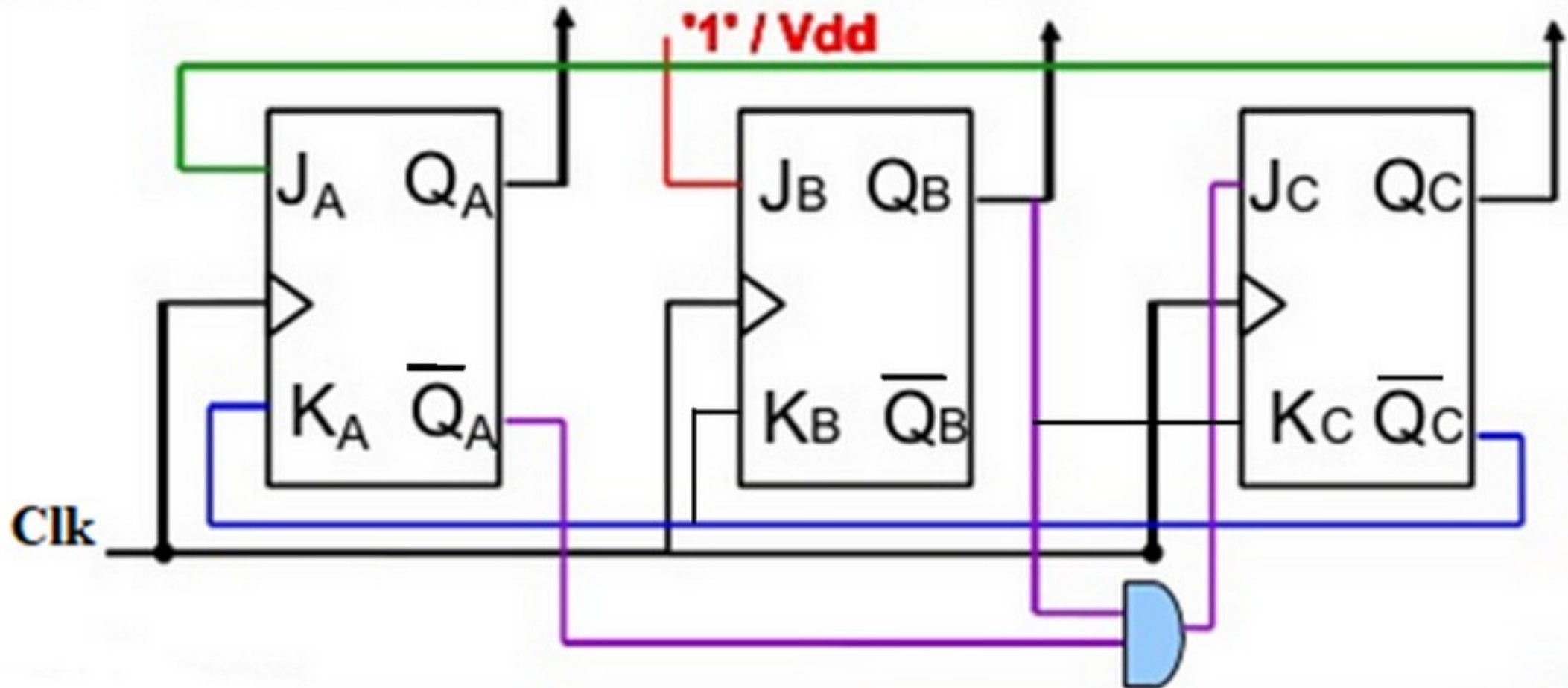
A \ CB	00	01	11	10
0	0	1	x	x
1	x	0	x	x

K-Map For
KC = QB

A \ CB	00	01	11	10
0	x	x	x	0
1	x	x	1	x

Step 5 : Perform Counter Circuit

By using simplified function from K-Map, **$J_A = Q_C$** , **$K_A = \overline{Q_C}$** ,
 $J_B = 1$, **$K_B = \overline{Q_C}$** , **$J_C = Q_A$** , **$K_C = Q_B$**



current state			next state			D _C	D _B	D _A
C	B	A	C*	B*	A*			
0	0	0	X	X	X	X	X	X
0	0	1	0	1	1	0	1	1
0	1	0	X	X	X	X	X	X
0	1	1	1	1	0	1	1	0
1	0	0	X	X	X	X	X	X
1	0	1	0	0	1	0	0	1
1	1	0	1	0	1	1	0	1
1	1	1	X	X	X	X	X	X

$$D_C = \Sigma m(3,6) + d(0,2,4,7)$$

$$D_B = \Sigma m(1,3) + d(0,2,4,7)$$

CB	00	01	11	10
A	X	X	1	X
0		1		
1			X	

CB	00	01	11	10
A	X	X		X
0				
1	1	1	X	

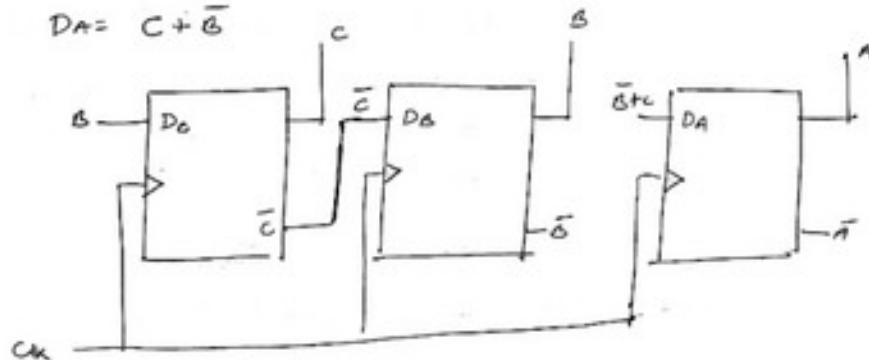
$$D_C = B$$

$$D_B = \bar{C}$$

$$D_A = \Sigma m(1,5,6) + d(0,2,4,7)$$

CB	00	01	11	10
A	X	X	1	X
0				
1	1		X	1

$$D_A = C + \bar{B}$$



This counter counts 1, 3, 6, 5, 1 using D flip flop.
To get the minimized cost the unused state can be assumed to go to don't care state

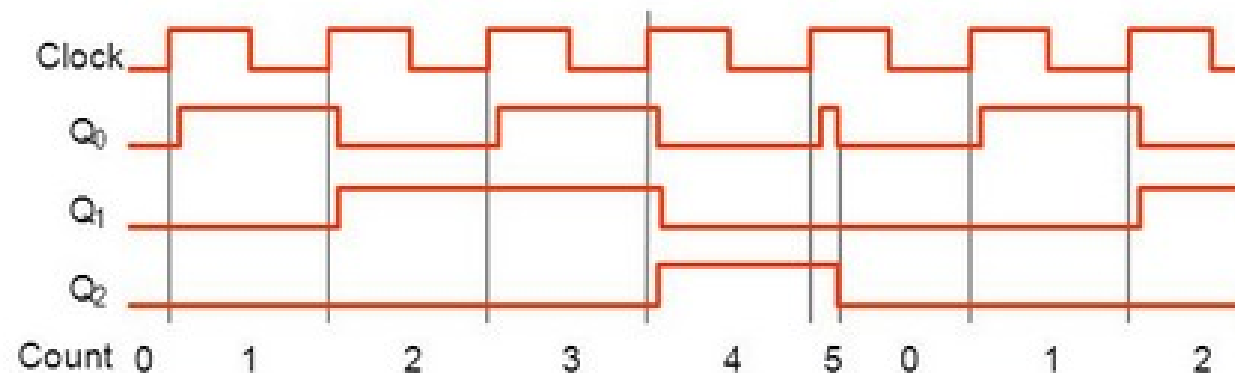
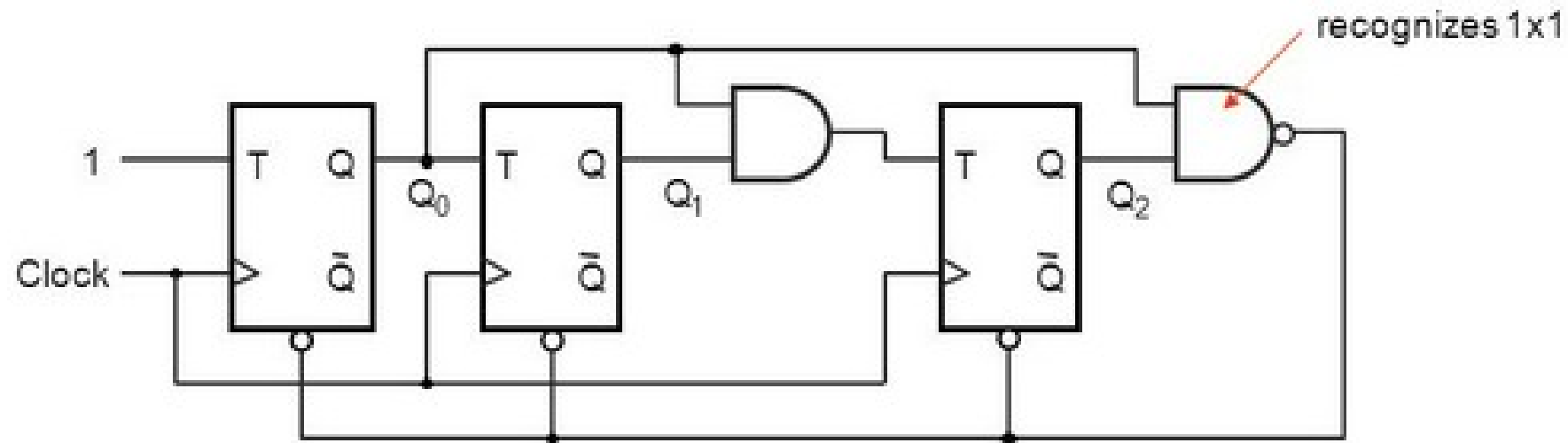
Q_n	Q_{n+1}	D
0	0	0
0	1	1
1	0	0
1	1	1

Use D flip-flop

Exercises:

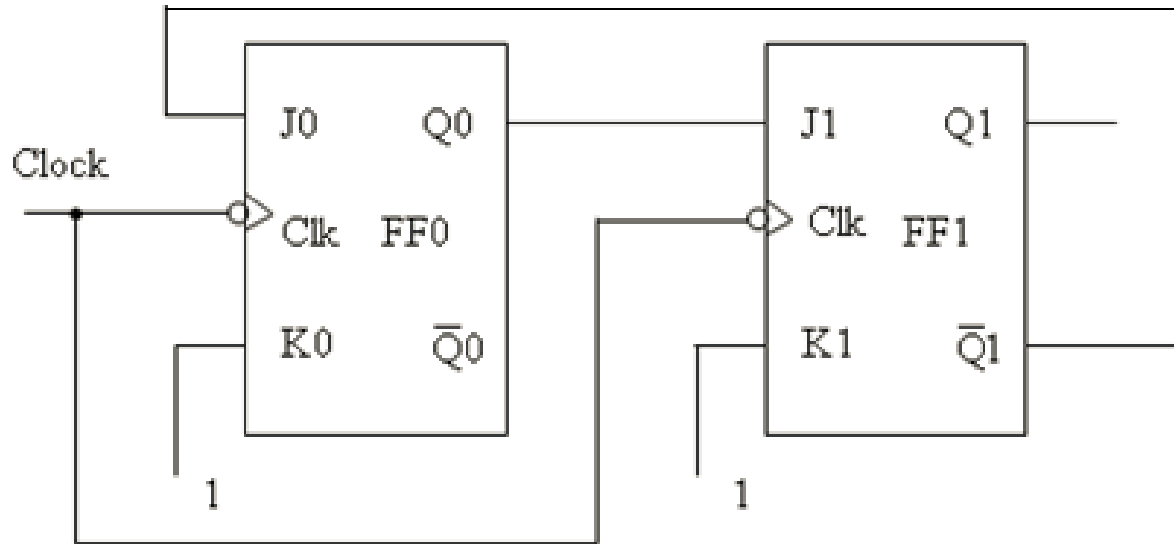
- **Design a counter to count in the following sequence: 6, 4, 2, 3, 1.**
- **Design a counter to count in the following sequence: 15,9,11,5,2,13,1.**

Identify the output of the circuit – if the clear signal is assumed to be asynchronous signal





Home Work - Identify the output of the following Circuit



Assume initial value on Q1 Q0 = 00

Clock	Q1	Q0
	0	0
1 st	0	1
2 nd	1	0
3 rd	0	0
4 th	0	1